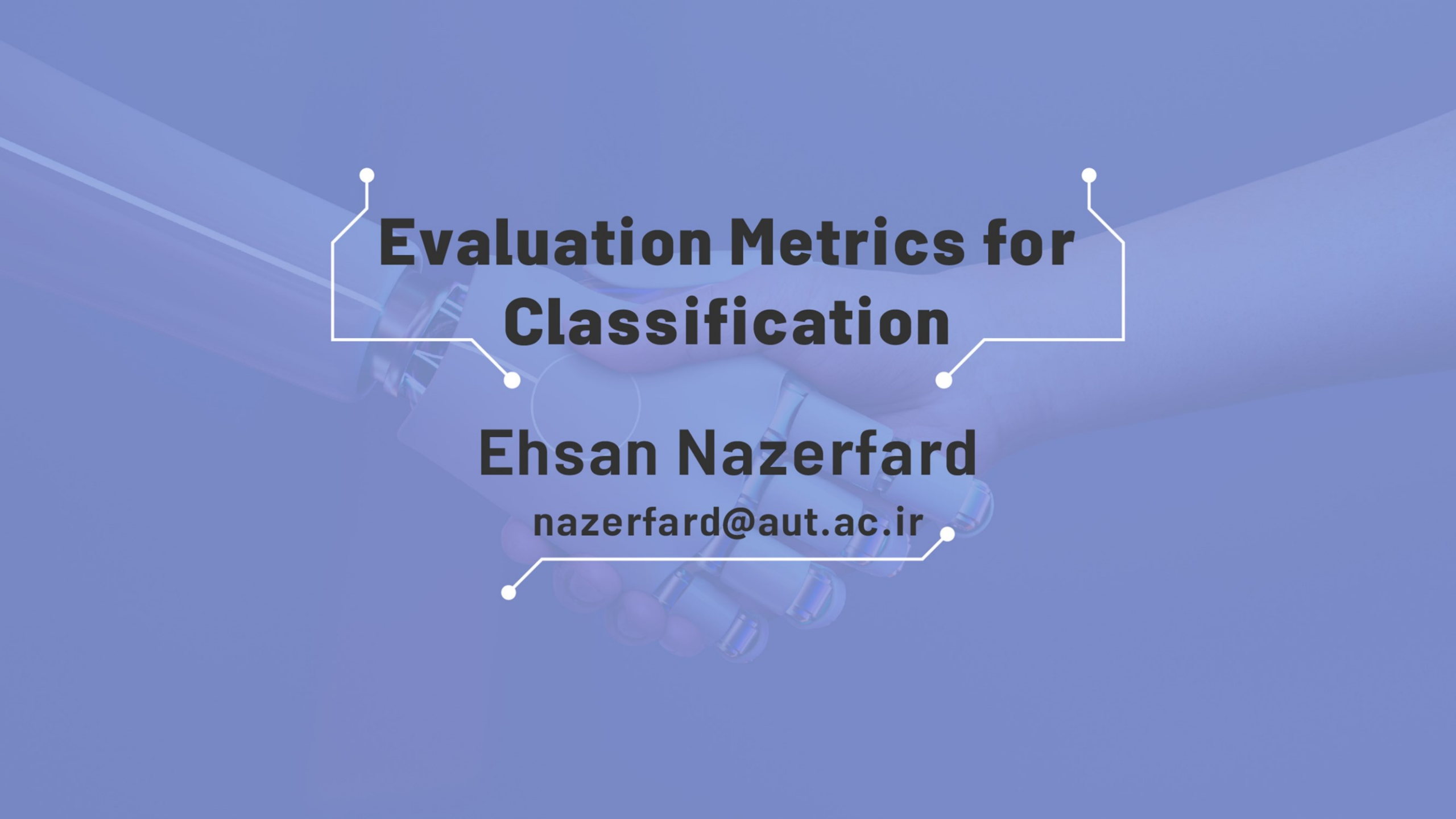




Evaluation Metrics for Classification

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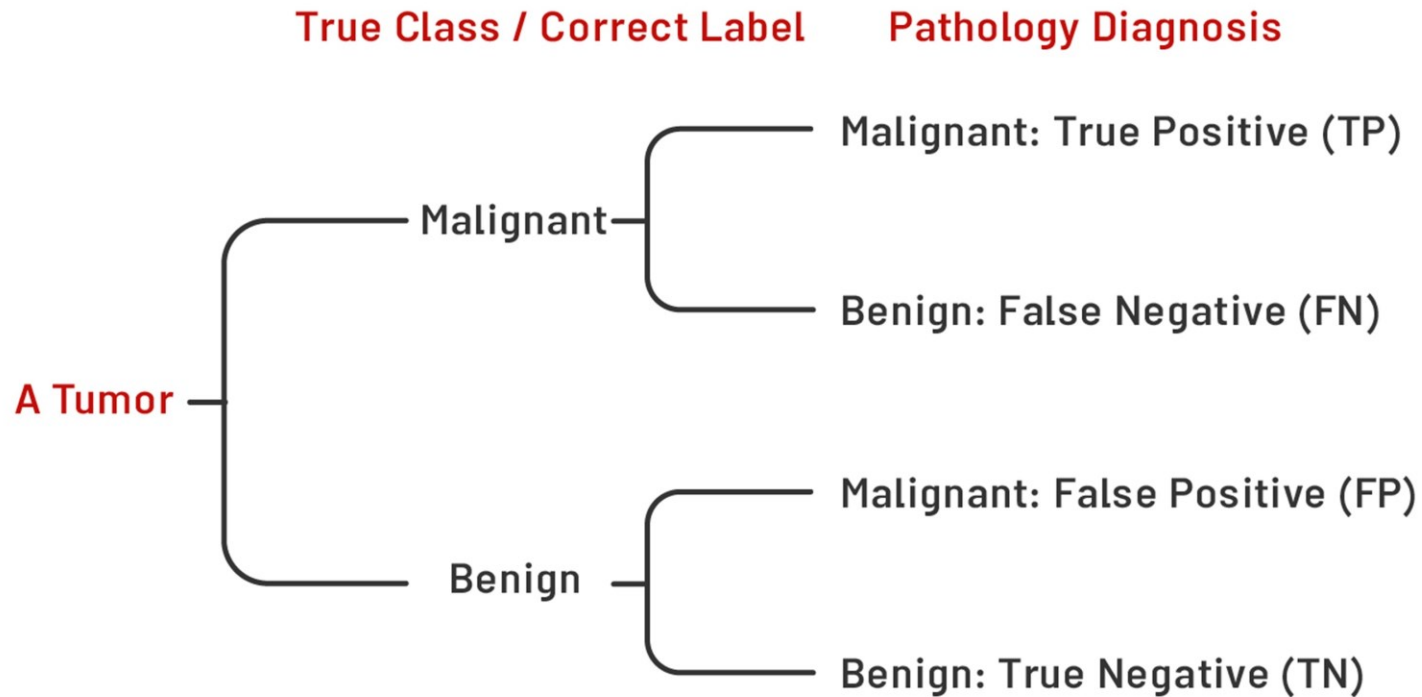


Class of Interest

- ▶ Spam Filtering → Spam (+), Legitimate (-)
- ▶ Tumor Diagnosis → Malignant (+), Benign (-)
- ▶ Activity Recognition in Smart Environments
- ...



Prediction Errors: Type I, II





Evaluation Metrics

$$\text{▶ } TP \text{ Rate} = TPR = \frac{\#TP}{\#P} = \frac{\#TP}{\#TP + \#FN}$$

$$\text{▶ } TN \text{ Rate} = TNR = \frac{\#TN}{\#N} = \frac{\#TN}{\#TN + \#FP}$$

▶ FPR & FNR are defined similarly.

$$\text{▶ } Accuracy = \frac{\#TP + \#TN}{\#P + \#N} = \frac{\#TP + \#TN}{\#TP + \#FN + \#TN + \#FP}$$

$$\text{▶ } Error \text{ Rate} = 1 - Accuracy$$



Evaluation Metrics (cont.)

▶ $TPR = \frac{TP}{TP + FN} = \text{Sensitivity} = \text{Recall}$

▶ how many relevant items are selected

▶ $TNR = \frac{TN}{TN + FP} = \text{Specificity}$

▶ $\text{Precision} = \frac{TP}{TP + FP}$

▶ how many selected items are relevant



Evaluation Metrics (cont.)

► $TPR + FNR = 1$

► $TNR + FPR = 1$

► $G - mean = (Precision \times Sensitivity)^{\frac{1}{2}}$

► $F_Score = 2 \times \frac{Precision \times Sensitivity}{Precision + Sensitivity}$



Confusion Matrix

► Confusion matrix for binary classification

predicted as			
correct label		Positive	Negative
	Positive	TP	FN
	Negative	FP	TN



Further Reading

- ▶ Imbalanced Learning
 - ▶ Cost sensitive learning
 - ▶ Oversampling/ undersampling techniques
 - ▶ SMOTE (Synthetic Minority Oversampling Technique)
- ▶ Receiver Operating Characteristic (ROC) Curves (later on in the course)
- ...